VU KHANH CAO

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EDUCATION

Carleton University, Ottawa, ON (GPA: 3.75/4.00) Sep 2023 – Apr 2028
Bachelor of Computer Science (Honours) – Artificial Intelligence and Machine Learning
Coursework: Object-Oriented Software Engineering, Distributed Computing, Design and Analysis of Algorithms

EXPERIENCE

Software Developer May 2026 – Present
Bank of Canada *Ottawa, ON*

- Modernized Treasury Management Systems processing $1T+ in annual transactions by developing **Java Spring Boot** services and TypeScript applications, optimizing performance to reduce load times by **80%** and query times by **35%**.
- Built and deployed an agentic code-review suite via **ArgoCD**, integrating with Azure CI/CD pipelines to automate security and code-quality checks and reduce security policy violations by **60%** across production repositories.
- Migrated legacy applications to **OpenShift** and containerized deployment environments, enabling scalable workloads and more consistent releases across systems supporting **2,000+ employees.**

Software Developer May 2025 – Dec 2025
Ovenns Analytics Inc. *Ottawa, ON*

- Developed a logistic regression classifier using **SciPy and NumPy** to categorize 3,000+ ingredients and preservatives into food groups, achieving 96% classification accuracy.
- Engineered a **Node.js and TypeScript** pipeline to parse online restaurant menus, normalize menu data, and decompose compound ingredients into basic ingredients, supporting onboarding for **100+ clients.**

PROJECTS

FLIH – Devpost Hack The North - Sep 2026

- Developed a **reinforcement learning pipeline** and MuJoCo simulation for a fly-connectome-based obstacle avoidance controller, reducing optimization step time from **8 minutes to 20 seconds** by minimizing CPU–GPU data transfers.
- Integrated **Huawei OMNI multimodal models** with onboard camera and LiDAR inputs for scene understanding and obstacle detection on Jetson, and built a React dashboard for robot control and training visualization over bidirectional WebSockets.

GURT – GitHub Jan 2026

- Developed a custom **QNX 8 camera driver in C**, increasing video throughput from **∼3 FPS to a stable 30 FPS (10×)** and earning **Best Hardware Hack** at uOttaHack 8.
- Engineered a **real-time robotic control stack** spanning QNX 8, Raspberry Pi 4, Arduino motor control, and WebSocket communication, enabling live video streaming with **sub-100 ms command latency**.

JARVIS – GitHub uOttaHack - Jan 2025

- Built a low-cost **physical-to-digital reconstruction pipeline** using QNX, Raspberry Pi, and Gaussian Splatting to produce web-ready 3D scenes from webcams, reducing capture hardware cost from **$400 to $20**.
- Containerized a Gaussian Splatting reconstruction pipeline and integrated **AWS S3** for image ingestion and generated-scene delivery, with GPU-accelerated processing for web-based 3D visualization.

TRACY – GitHub QHacks - Feb 2024

- Developed a **monocular computer vision pipeline** using TrackNetV2 and convolutional RNNs to track tennis ball and player trajectories from a single camera, eliminating the need for specialized tracking hardware.
- Built gameplay analytics for **serve speed, ball bounces, player heatmaps, and in/out detection** using Matplotlib and FFmpeg, contributing to a **3rd-place finish at QHacks 2024**.

SKILLS

Languages	Python, Java, JavaScript, TypeScript, C/C++, SQL
Libraries & Frameworks	React, Angular, Node.js, Express, Spring Boot, TensorFlow, OpenCV
Platforms & Technologies	Azure, AWS, Docker, Kubernetes, ArgoCD, OpenShift, REST APIs, Git